DESIGN AND ANALYSIS OF ALGORITHMS

LAB WORKBOOK

WEEK - 5

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CLASS : CSE-B

Question 1: Construct an AVL tree with these numbers:

157, 110, 147, 122, 149, 151, 111, 141, 112, 123, 133, 117

(i) Print the tree showing each level.

(ii) Check that all nodes are balanced.

CODE:

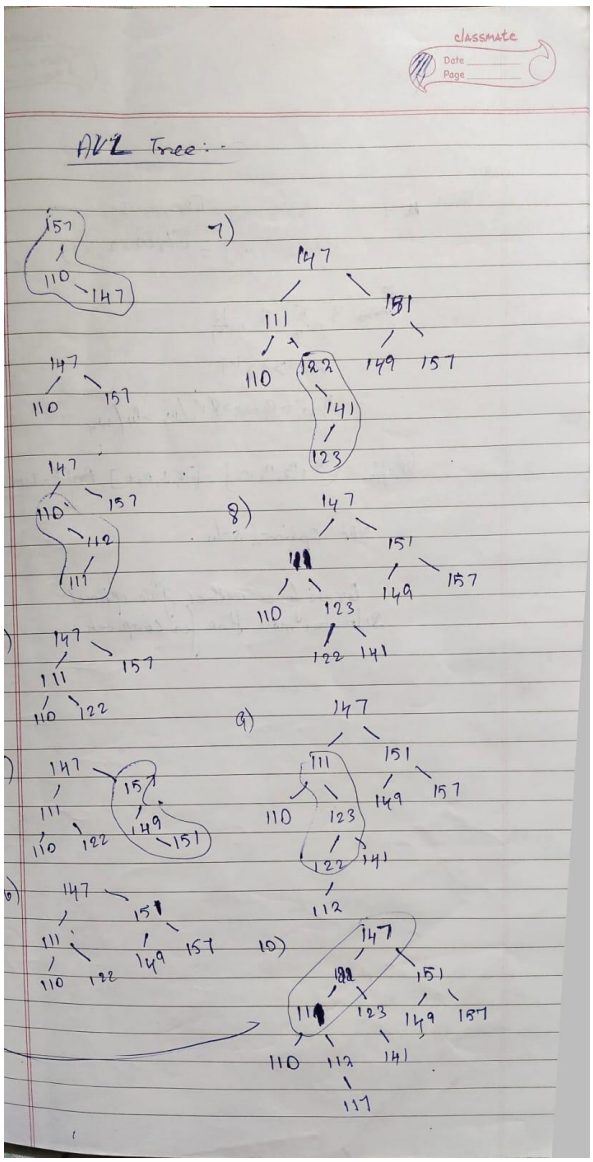
```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *left, *right;
    int height;
};
int max(int a, int b) {
    return (a > b) ? a : b;
}
int height(struct Node *n) {
    if (n == NULL)
        return 0;
    return n->height;
}
struct Node* newNode(int data) {
    struct Node* node = (struct Node*)malloc(sizeof(struct Node));
    node->data = data;
    node->left = node->right = NULL;
    node->height = 1;
    return node;
}
struct Node* rightRotate(struct Node* y) {
    struct Node* x = y->left;
    struct Node* T2 = x->right;
```

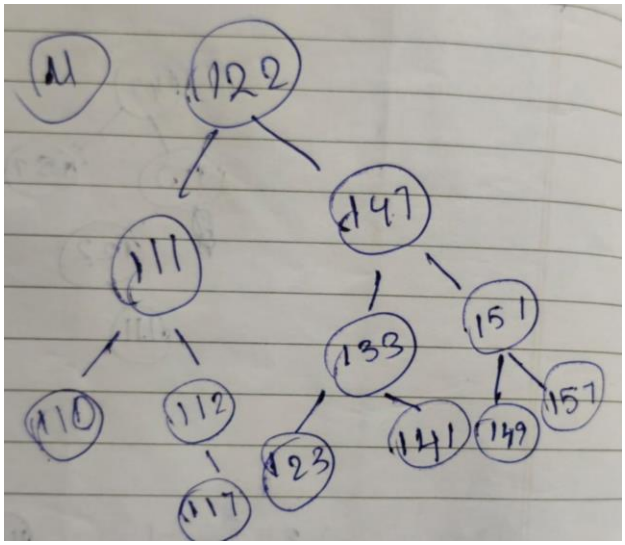
```
    x->right = y;
    y->left = T2;
    y->height = max(height(y->left), height(y->right)) + 1;
    x->height = max(height(x->left), height(x->right)) + 1;
    return x;
}
struct Node* leftRotate(struct Node* x) {
    struct Node* y = x->right;
    struct Node* T2 = y->left;
    y->left = x;
    x->right = T2;
    x->height = max(height(x->left), height(x->right)) + 1;
    y->height = max(height(y->left), height(y->right)) + 1;
    return y;
}
int getBalance(struct Node* n) {
    if (n == NULL)
        return 0;
    return height(n->left) - height(n->right);
}
struct Node* avlInsert(struct Node* node, int data) {
    if (node == NULL)
        return newNode(data);
    if (data < node->data)
        node->left = avlInsert(node->left, data);
    else if (data > node->data)
        node->right = avlInsert(node->right, data);
    else
        return node;
    node->height = 1 + max(height(node->left), height(node->right));
    int balance = getBalance(node);
    // Left Left Case
    if (balance > 1 && data < node->left->data)
```

```
        return rightRotate(node);
    // Right Right Case
    if (balance < -1 && data > node->right->data)
        return leftRotate(node);
    // Left Right Case
    if (balance > 1 && data > node->left->data) {
        node->left = leftRotate(node->left);
        return rightRotate(node);
    }
    // Right Left Case
    if (balance < -1 && data < node->right->data) {
        node->right = rightRotate(node->right);
        return leftRotate(node);
    }
    return node;
}
void levelOrder(struct Node* root) {
    if (root == NULL) {
        printf("Tree is empty\n");
        return;
    }
    struct Node* queue[100];
    int front = 0, rear = 0;
    int currentLevel = 0;
    queue[rear++] = root;
    printf("\nLevel Order Traversal:\n");
    while (front < rear) {
        int levelSize = rear - front;
        printf("Level %d: ", currentLevel++);
        for (int i = 0; i < levelSize; i++) {
            struct Node* current = queue[front++];
            printf("%d ", current->data);
            if (current->left != NULL)
```

```
                queue[rear++] = current->left;
            if (current->right != NULL)
                queue[rear++] = current->right;
        }
        printf("\n");
    }
}
void printTree(struct Node* root, int space) {
    if (root == NULL)
        return;
    space += 10;
    printTree(root->right, space);
    printf("\n");
    for (int i = 10; i < space; i++)
        printf(" ");
    printf("%d\n", root->data);
    printTree(root->left, space);
}
int main() {
    struct Node* root = NULL;
    int values[] = {157, 110, 147, 122, 149, 151, 111, 141, 112, 123, 133,
117};
    int n = sizeof(values)/sizeof(values[0]);
    printf("BUILDING AVL TREE     \n");
    printf("Inserting values: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", values[i]);
        root = avlInsert(root, values[i]);
    }
    printf("\n");
    levelOrder(root);
    if (getBalance(root) >= -1 && getBalance(root) <= 1)
        printf("Tree is AVL balanced\n");
```

```
    else
        printf("Tree is NOT balanced\n");
    return 0;
}
```





OUTPUT:

```
PS C:\Documents\sem4\daa\.vscode\week 5> gcc avltree.c
PS C:\Documents\sem4\daa\.vscode\week 5> ./a
Inserting values: 157 110 147 122 149 151 111 141 112 123 133 117

Level Order Traversal:
Level 0: 122
Level 1: 111 147
Level 2: 110 112 133 151
Level 3: 117 123 141 149 157
Tree is AVL balanced
PS C:\Documents\sem4\daa\.vscode\week 5>
```

Time Complexity for

(i) Search: O(log N) : Tree height is O(log n), search follows one path from root to leaf.

(ii) Insertion: O(log N) : BST insertion O(log n) + at most 2 rotations O(1) = O(log n).

(iii) Deletion: O (log N) : BST deletion O(log n) + rebalancing up to O(log n) rotations = O(log n)

(iv) Traversal: O(N) : Must visit all n nodes exactly once.

(v) Rotation: O(1) : Only changes a constant number of pointers.

Space Complexity: O(N) : Stores all the N nodes along with the data, pointer and height.

Question 2: Construct a Red-Black tree with the numbers:

157, 110, 147, 122, 149, 151, 111, 141, 112, 123, 133, 117

Print the tree showing R (red) or B (black) for each node. Check that:

(i) Root is black

(ii) No red node has red children

CODE:

```
#include <stdio.h>
#include <stdlib.h>
```

```c
#define RED 1
#define BLACK 0
struct Node {
    int data;
    int color;
    struct Node *left, *right, *parent;
};
struct Node *root = NULL;
struct Node* createNode(int data) {
    struct Node* node = (struct Node*)malloc(sizeof(struct Node));
    node->data = data;
    node->color = RED; // New node is always RED
    node->left = node->right = node->parent = NULL;
    return node;
}
void leftRotate(struct Node *x) {
    struct Node *y = x->right;
    x->right = y->left;
    if (y->left != NULL)
        y->left->parent = x;
    y->parent = x->parent;
    if (x->parent == NULL)
        root = y;
    else if (x == x->parent->left)
        x->parent->left = y;
    else
        x->parent->right = y;
    y->left = x;
    x->parent = y;
}
void rightRotate(struct Node *y) {
    struct Node *x = y->left;
    y->left = x->right;
    if (x->right != NULL)
```

```
        x->right->parent = y;
    x->parent = y->parent;
    if (y->parent == NULL)
        root = x;
    else if (y == y->parent->left)
        y->parent->left = x;
    else
        y->parent->right = x;
    x->right = y;
    y->parent = x;
}
void fixInsert(struct Node *z) {
    while (z != root && z->parent->color == RED) {
        if (z->parent == z->parent->parent->left) {
            struct Node *y = z->parent->parent->right; // Uncle
            // Case 1: Uncle is RED
            if (y != NULL && y->color == RED) {
                z->parent->color = BLACK;
                y->color = BLACK;
                z->parent->parent->color = RED;
                z = z->parent->parent;
            }
            // Case 2 & 3
            else {
                if (z == z->parent->right) {
                    z = z->parent;
                    leftRotate(z);
                }
                z->parent->color = BLACK;
                z->parent->parent->color = RED;
                rightRotate(z->parent->parent);
            }
        }
        else {
```

```
            struct Node *y = z->parent->parent->left;
            if (y != NULL && y->color == RED) {
                z->parent->color = BLACK;
                y->color = BLACK;
                z->parent->parent->color = RED;
                z = z->parent->parent;
            }
            else {
                if (z == z->parent->left) {
                    z = z->parent;
                    rightRotate(z);
                }
                z->parent->color = BLACK;
                z->parent->parent->color = RED;
                leftRotate(z->parent->parent);
            }
        }
    }
    root->color = BLACK;
}
void insert(int data) {
    struct Node *z = createNode(data);
    struct Node *y = NULL;
    struct Node *x = root;
    while (x != NULL) {
        y = x;
        if (z->data < x->data)
            x = x->left;
        else
            x = x->right;
    }
    z->parent = y;
    if (y == NULL)
        root = z;
```

```
    else if (z->data < y->data)
        y->left = z;
    else
        y->right = z;
    fixInsert(z);
}
int height(struct Node* node) {
    if (node == NULL) return 0;
    int leftHeight = height(node->left);
    int rightHeight = height(node->right);
    return (leftHeight > rightHeight ? leftHeight : rightHeight) + 1;
}
void printLevel(struct Node* node, int level, int currentLevel) {
    if (node == NULL) {
        if (currentLevel == level) printf("   ");
        return;
    }
    if (currentLevel == level) {
        printf("%d(%s) ", node->data, node->color == RED ? "R" : "B");
    } else if (currentLevel < level) {
        printLevel(node->left, level, currentLevel + 1);
        printLevel(node->right, level, currentLevel + 1);
    }
}
void levelOrder() {
    int h = height(root);
    printf("\n\nLevel Order Traversal:\n");
    for (int i = 0; i < h; i++) {
        printf("Level %d: ", i);
        printLevel(root, i, 0);
        printf("\n");
    }
}
void printTree(struct Node* root, int space, char* prefix) {
```

```c
    if (root == NULL) return;
    space += 10;
    printTree(root->right, space, "/");
    printf("\n");
    for (int i = 10; i < space; i++) printf(" ");
    printf("%s", prefix);
    printf("%d(%s)\n", root->data, root->color == RED ? "R" : "B");
    printTree(root->left, space, "\\");
}
void inorder(struct Node *node) {
    if (node != NULL) {
        inorder(node->left);
        printf("%d(%s) ", node->data,
               node->color == RED ? "R" : "B");
        inorder(node->right);
    }
}
int main() {
    int keys[] = {157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133};
    int n = sizeof(keys) / sizeof(keys[0]);
    printf("Inserting values in order:\n");
    for (int i = 0; i < n; i++) {
        printf("%d ", keys[i]);
        insert(keys[i]);
    }
    printf("\n\nInorder Traversal:\n");
    inorder(root);
    levelOrder();
    printf("\nTree Structure:\n");
    printf("===============\n");
    printTree(root, 0, "");
    return 0;
```

```
}
```

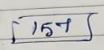
Red-Black Tree

Rules:

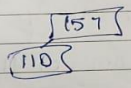
The nodes are either Red or Black
The root node is always Black
All NULL nodes are black
A Red node cant have Red parent/Child.
For every node all paths from the node to its descendant NULL leaves contains the same number of black nodes.

157, 110, 147, 122, 111, 149, 141, 183, 112, 117

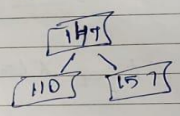
Step 1:



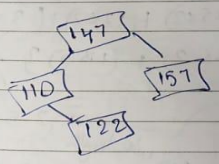
Step 2:



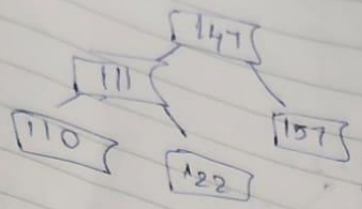
Step 3:



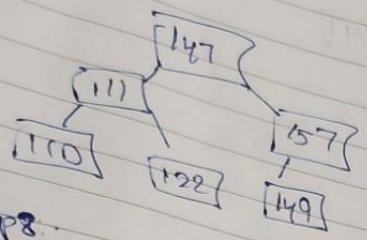
Step 4:



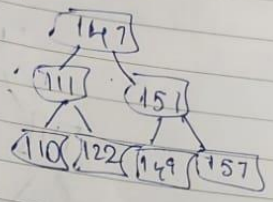
Step 5:-



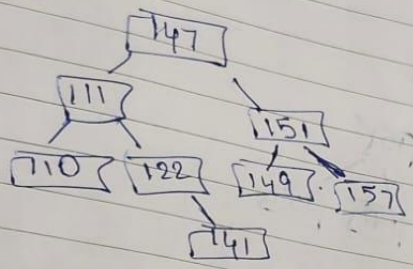
Step 6:-



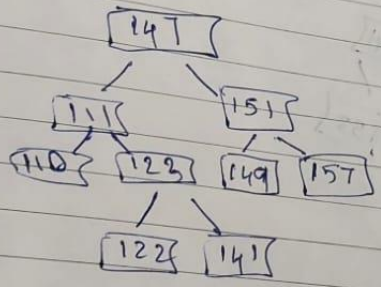
Step 7:- 151



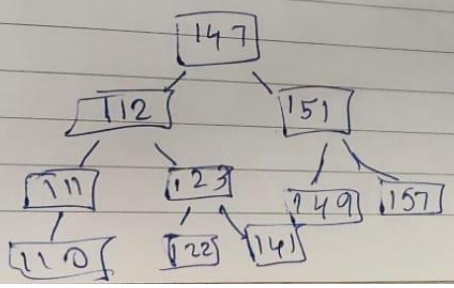
Step 8:-

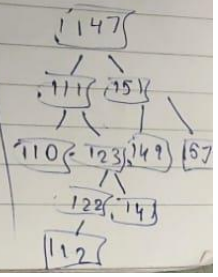


Step 9:-



Step 10:-





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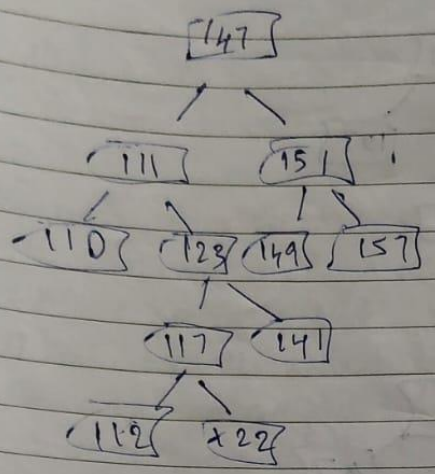
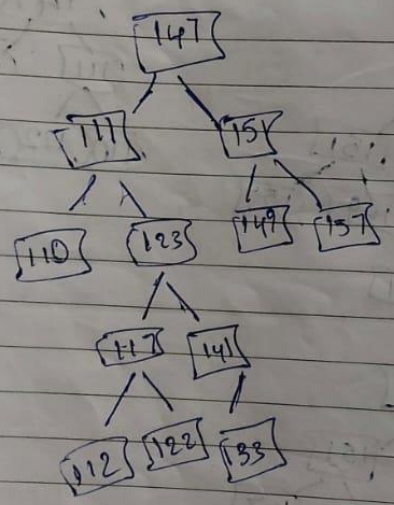
Step 11:



Step 12: 133



OUTPUT:

```
Inserting values in order:
157 110 147 122 111 149 151 141 123 112 117 133

Inorder Traversal:
110(B) 111(R) 112(R) 117(B) 122(R) 123(B) 133(R) 141(B) 147(R) 149(R) 151(B) 157(R)

Level Order Traversal:
Level 0: 123(B)
Level 1: 111(R) 147(R)
Level 2: 110(B) 117(B) 141(B) 151(B)
Level 3:       112(R) 122(R) 133(R)    149(R) 157(R)
```

Time Complexity for

(i) Search: O(log N) : Red-Black properties guarantee tree height $\leq 2*\log_2(n+1)$, so we traverse at most logarithmic levels from root to leaf.

(ii) Insertion: O(log N) : BST insertion takes O(log n) to find the position, and fixing violations requires at most O(log n) recoloring plus at most 2 rotations (O(1) each).

(iii) Deletion: O (log N) : BST deletion takes O(log n) to find the node, and fixing violations requires at most O(log n) recoloring plus at most 3 rotations (O(1) each).

(iv) Traversal: O(N) : We must visit every node exactly once in inorder /preorder/ postorder traversal, and there are n nodes.

(v) Rotation: O(1) : It only updates a constant number of pointers (parent, left, right) regardless of tree size.

Space Complexity: O(N) : Stores all the N nodes along with the data, pointer and colour information.